

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech 4th Semester Mid-Semester Examination,
February 2026.
Design and Analysis of Algorithms (CS2201)

Full Marks: 30

Time: 2 Hours

Answer any three(3) questions. It is advised to answer all parts of a question together. Credit will be given to precise answer.

1. (a) Suppose that for inputs of size n on a particular computer, insertion sort runs in $8n^2$ steps and merge sort runs in $64n \lg n$ steps. For which values of n does insertion sort beat merge sort?
(b) Prove that $2n^2 - 4n + 7 = \Theta(n^2)$.
(c) Prove that you can not find the minimum elements in an array of n elements in lesser than $O(n)$ time. [2+4+4]
2. (a) Propose an efficient algorithm to find the minimum and second minimum elements in an array of n elements, also find out the running time of your proposed algorithm.
(b) What is merging? What is the running time of merging two sorted arrays of n elements each and provide a proof of the estimated running time.
(c) In what way merge sort is different from quick sort? [4+4+2]
3. (a) What do you understand by approximate median in an array of n elements. Propose an efficient algorithm to find the median in an array of n elements, hence prove the running time of your proposed method.
(b) What would be the running time of the quick sort algorithm if it always produces
(i) 9:1 proportional split?
(ii) good ($n/2$ elements in each half) partitioning and bad ($1 : n - 1$) partitioning alternatively? [6+4]
4. (a) You are given a set of intervals in the form of their starting time and ending time. Propose an algorithm to find out the maximum number of non-overlapping intervals and give a proof of correctness of your proposed algorithm.
(b) What do you understand by maximal independent set in a graph? Prove that the leaf nodes must always fall in some optimum Independent Set in a tree. [6+4]